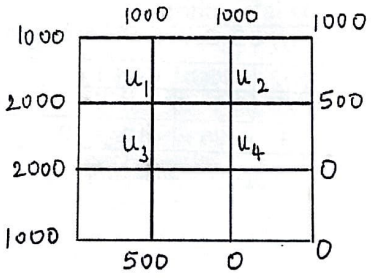
CIE - II

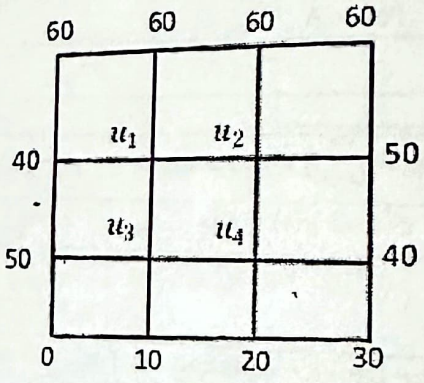
RV College of Engineering

DEPARTMENT OF MATHEMATICS

Academic year 2024-2025 (Odd Semester 2024)

Date	**02.12.2024**	**Time**	**9.30 AM– 11.30 AM**
TEST	**Quiz- II & Test -II**	**Maximum Marks**	**10+50**
Course Title	**STATISTICS, LAPLACE TRANSFORM AND NUMERICAL METHODS**	**Course Code**	**MA231TB**
Semester	**III**	**Programs**	**AS, BT, CH, ME, IEM**

Sl. No	Instructions: Answer all questions. Part - A	M	CO	BT
1	$L\left[\int_0^t \cosh t\right] =$ _____.	2	1	L1
2	If $f(t) = te^{-2t}$ then $L[f(t)\delta(t-4)] =$ _____.	1	1	L1
3	The inverse Laplace transform of $F(s) = \frac{s}{(s+a)^2}$ is _____.	2	1	L2
4	The convolution of functions $f(t) = e^t$ and $g(t) = t$ is _____.	2	2	L2
5	$L^{-1}\left[\frac{2}{(s-1)^2+4}\right] =$ _____.	1	1	L1
6	The initial approximation of u_3 in the Laplace equation $u_{xx} + u_{yy} = 0$ at the pivotal points of the given figure using standard five point formula and assuming $u_4 = 0$ is _____. 	2	1	L2
	Part - B			
1a	Determine the Laplace transform of the periodic function defined by $f(t) = \begin{cases} 1, & 0 \le t < a \\ -1, & a < t \le 2a \end{cases}$ and $f(2a+t) = f(a)$.	4	2	L2
1b	Express the following function in terms of unit step function and hence obtain its Laplace transform $f(t) = \begin{cases} t^2, & 1 < t \le 2 \\ 4t, & t > 2 \end{cases}$.	6	3	L3
2	Obtain the inverse Laplace transform of the function $F(s) = \frac{s}{(s+1)(s^2+1)}$ using the method of partial fractions.	10	2	L2
3	Employing convolution theorem, determine the inverse Laplace transform of $F(s) = \frac{s^2}{s^4-16}$.	10	3	L3

4	The displacement $x(t)$ of a mass-spring system is governed by $\frac{d^2x}{dt^2}+\lambda\frac{dx}{dt}+k\,x=f(t)$, where λ and k are force constants subject to the conditions $x(0)=2, \frac{dx}{dt}(0)=-4$ and $\lambda=2, k=5, f(t)=\frac{17}{2}\sin 2t$. Determine $x(t)$ by using Laplace transform method.	10	4
5	Given the values of $u(x,y)$ on the boundary of the square region. Evaluate the function $u(x,y)$ satisfying the Laplace equation $u_{xx}+u_{yy}=0$ at the pivotal points of the figure. Perform four iterations. 	10	3

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BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5
	Quiz Max Marks	8	2			4	6			
	Test Max Marks	0	14	26	10	0	14	26	10	

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Date	3
Course Code	C
Sem	
Enviro	

Sl. No.	
1.	Describe the t
2.	Explain the n
3.	Classify diffe
4.	Differentiate systems.
5.	Describe th renewable e
6.	State the si
7.	Enumerate

Sl. No.	
1.	a. Illustra b. What
2.	Describ limitati
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Course outc

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Department of Civil Engineering

Date	3/12/2024	Maximum Marks	60
Course Code	CV232AT	Duration	90 Min +20 Min
Sem	III	CIE – II (Quiz + Test)	
Environment and Sustainability (Basket Course)			

Part A

Sl. No.	Questions	M	BT	CO
1.	Describe the term of "environmental audit"?	1	1	2
2.	Explain the need for new energy sources.	1	2	3
3.	Classify different types of energy sources.	1	2	3
4.	Differentiate between linear and cyclical resource management systems.	1	2	3
5.	Describe the working principle of tidal energy converter in renewable energy generation.	2	1	2
6.	State the significance of GDP in environmental economics.	2	1	2
7.	Enumerate the importance of environmental economics.	2	2	3

Part B

Sl. No.	Questions	M	BT	CO
1.	a. Illustrate carbon cycle with sketch. b. What are the advantages of hydrogen energy?	05 05	2 2	2 2
2.	Describe ocean thermal energy conversion with its advantages and limitations.	10	2	2
3.	Enumerate the concept of sustainable urbanization and the need for the same.	10	2	3
4.	a. Discuss the steps in environmental audit. b. Enumerate the economics and social aspects of sustainability.	05 05	2 2	3 3
5.	Illustrate the concept of economics and its integration to social aspects for sustainability.	10	2	3

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	25	35	-	5	53	2	-	-	-

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Department of Mechanical Engineering

CIE – II

Date	3rd December 2024	Maximum Marks	50+10
Course Code	ME232TB	Duration	120 Min
Course Name	Material Science for Engineers	USN:	

Part A (Quiz)

Sl. No.	Questions	M	BT	CO
1.	Semiconductor device with a p-n junction which produces current by absorption of light in the depletion region is termed as _____	1	L1	2
2.	_____ is the metal joining process in which parent metal pieces are heated either to plastic or molten state.	1	L1	2
3.	If the emission of radiation occurs longer than 10^{-8} seconds is known as_____	1	L1	2
4.	Alloying elements of HSS are _____	1	L1	2
5.	Lightest of commonly used metals is _____	1	L1	2
6.	Define young's modulus of a material with equation.	2	L2	2
7.	_____ is the type of glass that cannot allow heat through it because of air between the layers and acts as good insulators.	1	L1	2
8.	Concrete is a composite material made up of _____ and _____	2	L1	2

Part-B (Test)

Sl. No.	Questions	M	BT	CO
1 a)	What is piezoelectric effect? Explain the applications of piezoelectric materials.	05	L2	2
b)	Distinguish between ductile and brittle fracture.	05	L2	2
2	With the stress strain diagram explain the mechanical properties of a ductile metal.	10	L2	2
3 a)	Discuss the applications of semiconductors.	05	L3	2
b)	Explain four different types of cast iron based on microstructure, properties and applications.	05	L2	2
4 a)	With a neat sketch describe the steps to be followed in metal casting process.	05	L2	2
b)	Compare thermoplastics with thermosetting plastics.	05	L3	2
5 a)	Justify the need for composite materials. Classify composite materials.	05	L3	2
b)	What is a biomaterial? List the desirable properties of biomaterials.	05	L2	2

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DEPARTMENT OF
BIOTECHNOLOGY
Academic year 2024-2025 (Odd Sem)

Date	December 3, 2024	Maximum Marks	10 + 50
Course Code	BT232AT	Duration	120 min
Sem	III Semester	Closed Book Offline Quiz and Test – 2	

Bio Safety Standards and Ethics (Basket course)

Answer all questions in Part A & Part B, Answer Part A in first two pages in sequence only

Part A

Sl No	Questions	M	BT-L	CO
1	A passenger in flight found a bolt and nut in the burger. Identify the type of the hazard.	1	2	3
2	Expand the term GMP	1	2	4
3	Name any one food certification followed in the world	1	2	3
4	Expand the term GHP	1	2	4
5	A person found a frog leg in the food. Identify the type of the hazard	1	2	4
6	_____ types of licences issued by FSSAI	1	2	3
7	Name two diseases caused due to food contamination	1	2	4
8	Milk is heated to _____ ^{0}C temperature during pasteurization	1	1	3
9	What is 2-hour 4-hour rule for food storage	2	2	3

Part B

Sl. No	Questions (Test)	M	BT-L	CO
1.	Define food contamination. Mention and explain with examples the types of Hazard which contaminate food	10	2	3
2	Elaborate on the basic rules of hygiene, sanitation and safety in food processing.	10	3	3
3	Explain in detail how to eliminate the different type of hazards which contaminate food.	10	3	3
4a	Mention the principles of HACCP	5	3	4
4b	 Decode the above FSSAI license number	5	2	3
5a	Explain in detail the instruments used in food testing laboratory	5	3	4
5b	Illustrate the various analytical methods used in food testing process.	5	3	4

BT-L-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	---	--	41	19	---	24	36	---	---	--

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USN										

Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	2nd December 2024	Maximum Marks	10 + 50
Course Code	IM233AI	Duration	120 Min
Sem	III Semester	CIE –II	

WORK SYSTEMS DESIGN

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	List any four techniques of Work Measurement.	02	1	1
2	What is meant by the term, " Muda" in the context of Lean Manufacturing?	02	2	2
3	Distinguish between work element and basic motion element.	02	2	1
4	What is the expanded form of MOST?	02	1	3
5	Mention the advocate of Toyota Production System.	02	1	2
	Part –B			
1	Illustrate and explain the pre requisites for determining the valid time standards.	10	3	3
2	A time study analyst observed a repetitive work cycle performed by a worker and then studied the task using a predetermined motion time system. During the observation, the analyst made a record of the work elements. He also made a note of worker's performance and rated it as 85%.Back in the office; the analyst listed the basic motion elements for the task and retrieved from his computer the corresponding normal time values. The sum of normal times for the basic motion elements was 1.38 min. The company uses a PFD allowance factor of 15%.Determine the standard time for the task.	10	3	3
3	The continuous timing method is used to direct time study a manual task cycle consisting of four elements: *a, b, c* and *d*. Two parts are produced each cycle. Element *d* is irregular element performed once in every six cycles. All elements were performance rated at 90%.The PFD allowance is 11%.Determine the normalized time for the cycle, standard time per part, the workers efficiency if the worker completes 844 parts in an 8-hour shift during which she works 7 hours and 10 minutes.	10	4	3
4	Enumerate different categories of Allowances considered in determining the time standards. Present your understanding by providing relevant contexts in which the different categories are applied by organizations in determining standard time.	10	4	1
5	Sketch the structure of Lean Production System. Review the key elements of the structure.	10	3	2

Element	a	b	c	d
Observed time (min)	0.35	0.60	0.86	1.46

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	06	02	02	10	06	04	--	--	--	--
	Test		--	--	30	20	10	20	20	--	--	--

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Academic year 2024-2025 (ODD Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**3rd December 2024**	**Maximum Marks**	**10 + 50**
Course Code	**IM234AI**	**Duration**	**120 Min**
Sem	**III Semester**	**CIE –II**	

MANUFACTURING PROCESSES

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	The _____ is the sharp edge at the point where the side cutting edge and end cutting edge meet	1	L1	CO2
2	A common defect in arc welding is _____, which occurs when the arc is unstable and leads to uneven welding	1	L2	CO3
3	The _____ casting process involves using a permanent metal mold and injecting molten metal under high pressure to create parts	1	L2	CO2
4	The _____ of a single-point tool is the angle formed between the flank of the tool and the workpiece surface, which helps in controlling the chip flow	1	L1	CO1
5	List any two applications of shell molding process	2	L2	CO2
6	List any two advantages of centrifugal casting	2	L1	CO2
7	State any two difference between orthogonal and oblique cutting	2	L1	CO3
	Part –B			
1	With a neat sketch explain the casting process which is best suited for casting of hollow pipes and tubes.	10	L2	CO1
2	With the help of neat sketch, explain the working principle of Projection Welding	10	L1	CO2
3	Explain the working principle of the welding process that uses a focused beam of high-velocity electrons to generate heat for melting and fusing materials, particularly in applications requiring precision, such as aerospace, automotive, and electronics manufacturing?	10	L3	CO3
4	Describe the working principle of the casting method that uses a wax pattern coated with a ceramic shell to create a mold, which is then used to produce highly detailed and complex components for applications such as aerospace turbine blades, medical implants, and precision automotive parts?	10	L3	CO2
5	What are the different types of chips that are formed in metal machining and give the conditions that are favorable to their formation?	10	L2	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	03	05	02	10	04	06	--	--	--	--
	Test		10	20	20	--	10	20	20	--	--	--

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Academic year 2024 – 2025 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING& MANAGEMENT

Date	4th December 2024	**Maximum Marks**	10 + 50
Course Code	IM235AI	**Duration**	120 Min
Sem	III Semester	CIE – II	

DIGITAL METROLOGY

Part – A

Sl. No.	Questions	M	BT	CO
1.	List the Basic components used in ADCs and DACs	02	L1	CO2
2.	Mention the principle of working of Piezoelectric sensor.	01	L2	CO2
3.	Distinguish between Unilateral and Bilateral Tolerance	02	L2	CO3
4.	How a Feature Control Frame represented in a GD&T technical drawing?	01	L1	CO3
5.	Differentiate between allowance and clearance?	02	L2	CO3
6.	What is Interchangeability?	02	L2	CO3

Part – B

Sl. No.	Questions	M	BT	CO
1.	Evaluate the operational mechanisms of tactile sensors critically analyzing their fundamental principles and contact detection capabilities. Provide examples of applications where these sensors are commonly used.	(10)	L3	CO2
2.	Determine the actual dimension to be provided for a shaft and hole of 70 mm size for F_8e_9 type of fit. i. Size 70 mm falls in the diameter step of 50mm – 80mm. ii. Value of tolerances for IT8 and IT9 are 25i and 40i. iii. Value of fundamental deviation for 'F' type shaft is $5.5D^{0.41}$ and for 'e' type shaft is $-11D^{0.41}$ Calculate the maximum and minimum clearance. Also mention the type of fit. Represent the dimensions graphically.	(10)	L4	CO3
3.	Discuss the different types of fits commonly used in engineering assemblies. Explain the characteristics, advantages, and applications of each type of fit, illustrating your answer with relevant examples.	(10)	L3	CO3
4.	Write a note Upper deviation and Lower deviation and Fundamental deviation. Describe the concepts of the hole basis system and shaft basis system in engineering fits. Explain how these systems are implemented in manufacturing processes and their significance in ensuring functional compatibility and quality of assembled parts.	(10)	L2	CO3
5.	Write a note on the following and represent schematically. Basic Hole, Basic Shaft, Minimum Limit of Size, Maximum clearance, Minimum Clearance	(10)	L2	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	--	03	07	--	--	03	07	--	--	--	--
	Test		--	10	40	--	--	--	20	20	10	--	--

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CSE-II - 2024

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Department of Mathematics

Course: Statistics, Laplace transform and Numerical Methods	**Test II**	**Maximum marks: 50**
Course code: MA231TB	**Third semester 2023-2024 (AS, BT, CH, IM, ME)**	**Time: 10 AM to 11:30 AM Date: 20/02/2024-**

Instructions to candidates: ***Answer all questions.***

Q.No	Questions	M	BTL	CO
1a	By using the convolution theorem evaluate: $L^{-1}[\frac{1}{(s+1)\,s^2}]$	5	1	1
1b	Solve $x''(t) + 9x(t) = \cos(2t)$, $x(0) = 1$, $x'(0) = \frac{12}{5}$ by method of Laplace transforms.	5	3	2
2 a	Evaluate: $L^{-1}[\frac{s\,e^{-\frac{s}{2}}+\pi\,e^{-s}}{s^2+\pi^2}]$	5	2	1
2 b	The equations of regression lines of two variables x and y are $x - y = 0$, $4x - y = 3$ and the standard deviation of x is 1. Obtain the variance of y and the coefficient of correlation between x and y.	5	2	2
3	Compute the first four moments about the mean of the frequency distribution given below:	10	2	3
4a	The following table gives ranking of 10 students according to their achievements in both the laboratory and lecture portions of an engineering course. Find the coefficient of rank correlation.	4	1	2
4b	If the velocity V (km/hr) and the resistance R $(kg/tonne)$ are related by a relation of the form $R = a + bV^2$, find a and b, by the method of least squares, given:	6	3	1
5	The following table shows the weights (z), heights (x) and ages (y) of 8 boys. Find the regression equation of z on x and y, estimate the weight of a boy who is 9 year old and 54 inches tall.	10	3	4

x	0	1	2	3	4
f	1	8	28	56	70

Lab(x)	8	3	9	2	7	10	4	6	1	5
Lecture(y)	9	5	10	1	8	7	3	7	2	6

V	10	20	30	40	50
R	8	10	15	21	30

z (lbs.)	64	71	53	67	55	58	77	57
x (in.)	57	59	49	62	51	50	55	48
y (yrs.)	8	10	6	11	8	7	10	9

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Test	Max Marks	16	14	10	10	9	20	21	--	--	--

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Department of Civil Engineering

Academic year 2023-2024 (Odd Sem)

Date	20-02-2024	Maximum Marks	50
Course Code	CV232AT	Duration	90 Min
Sem	III (Basket Course)	CIE – II (Test)	
ENVIRONMENT AND SUSTAINABILITY			

Sl. No.		Questions	Marks	BT	CO
1.		Enumerate the process of energy management and energy conservation with its objectives.	10	2	3
2.		Explain any 5 types of alternate energy sources with their advantages and disadvantages.	10	3	1
3.		Enumerate the objectives of sustainable urban projects.	10	2	3
4.		Highlight five important principles of sustainable development	10	3	3
5.	a.	Briefly explain any 5 international protocols and agreements that address specific environmental issues.	05	2	2
	b.	Describe any 5 disadvantages of linear resource management systems.	05	3	2

Course outcomes:

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	10	30	-	-	25	25	-	-	-

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Academic year 2023-2024 (Odd Sem)

DEPARTMENT OF
MECHANICAL ENGINEERING
CIE II

Date	**20th February 2024**	Maximum Marks	**50**
Course Code	**ME232AT**	Duration	**90 Minutes**
Course Name	**MATERIALS SCIENCE FOR ENGINEERS**		Sem: 3

Q. No.	Questions	M	BT	CO
1a.	With the help of a stress-strain diagram explain toughness and ductility of a material	04	3	2
1b.	Using stress-strain diagram distinguish ductile and brittle fracture	06	2	2
2.	Explain the following properties: thermal expansion coefficient, thermocouple. luminescence, optical fibres	10	3	2
3.	Describe the following: ferroelectricity, piezoelectricity, superconductor,	10	4	2
4.	Explain the properties and applications of following materials with examples. Ceramics and Fiber reinforced composites	10	3	2
5.	Explain the properties and applications of following materials. stainless steel, cast iron and titanium alloys.	10	3	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	TEST	Marks	-	50	-	-	06	30	10	4		

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Academic year 2023-2024 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**22nd February 2024**	**Maximum Marks**	**50**
Course Code	**IM234AI**	**Duration**	**90 Min**
Sem	**III Semester**	**CIE – II**	

MANUFACTURING PROCESS

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	Enumerate the importance of pattern making and discuss different types of pattern making materials.	10	1	1
2	List and discuss the essential properties required for a moulding sand.	10	2	2
3	Classify and explain the basic bulk forming process with a neat sketch.	10	3	3
4	Discuss the importance of roll presses with the following: i. Breakdown Passes ii. Roughing Passes iii. Finishing Passes	10	2	2
5	Differentiate with a neat sketch, between direct and indirect extrusion with the reference to following: i. Material flow ii. Force required iii. Another name iv. Material wastages v. Frictional force	10	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	20	--	10	20	20	--	--	--

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Academic year 2023-2024 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**24th February 2024**	**Maximum Marks**	**50**
Course Code	**IM235AI**	**Duration**	**90 Min**
Sem	**III Semester**	**CIE– II**	

DIGITAL METROLOGY

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	Explain the principle of Interference and Describe with neat sketches the working of an autocollimator. Explain how these tools are used in manufacturing processes for measurement of straightness.	(10)	L3	CO3
2.	Define Crest, Flank and Pitch and schematically represent on a screw thread. Explain how the major diameter of a screw thread can be measured using Bench micrometer.	(10)	L3	CO3
3.	Determine the actual dimension to be provided for a shaft and hole of 90 mm size for h_8E_9 type of fit. i. Size 90 mm falls in the diameter step of 80mm-100mm. ii. Value of tolerances for IT8 and IT9 are 25i and 40i. iii. Value of fundamental deviation for 'e' type shaft is $11D^{0.41}$. Also mention the type of fit. Represent the dimensions graphically.	(10)	L4	CO2
4.	Define Upper deviation and Lower deviation and Fundamental deviation. Describe the concepts of the hole basis system and shaft basis system in engineering fits. Explain how these systems are implemented in manufacturing processes and their significance in ensuring functional compatibility and quality of assembled parts.	(10)	L2	CO2
5.	Discuss different types of surface finish analysis techniques and irregularities commonly encountered in engineering. Explain the significance of surface finish analysis in manufacturing processes, and how identifying irregularities contributes to product quality and performance. Provide examples to support your explanation.	(10)	L3	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	--	20	30	--	--	--	10	30	10	--	--

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CIE II - 2023

DEPARTMENT OF MATHEMATICS

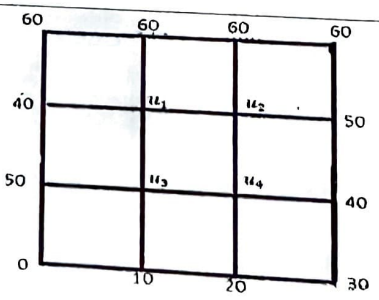
Date	20/02/2023	**Maximum Marks**	10+50
Course Code	21MA31C	**Duration**	120 Min
Semester	III	**CIE-II**	

Integral Transforms, Optimization, and Numerical Techniques

(BT, CH, CV, IEM, ME)

Instructions to candidates:

Answer all questions from part A and part B. Part A questions should be answered in the first two pages of the answer book only.

Sl No	PART-A	M	BT	CO
01	The diagonal five-point formula to evaluate $u_{2,2}$ is_____	1	2	1
02	The partial differential equation $u_{xx} + xu_{yy} = 0$ is classified as _____	1	1	2
03	For the one-dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, with the boundary conditions $u(0,t) = u(5,t) = 0$, and the initial conditions $u(x,0) = x(x-5)$ and $u_t(x,0) = 0$ with $h = 1, k = 1$, the value of $u(1,1)$ is _____	2	2	2
04	The corner points of the feasible region for the objective function $z = 3x + 2y$ subject to the constraints $x + 2y \leq 10,\ 3x + y \leq 15$ and $x, y \geq 0$ are _____	2	2	3
05	Write the following L.P.P. in the standard form: Maximize $Z = 3x_1 + 2x_2$ subject to $x_1 + 2x_2 \geq 4,\ x_1 - x_2 \leq 2,\ 2x_1 + 3x_2 = 6,\ x_1,\ x_2 \geq 0$.	2	2	1
06	Determine all the basic solutions of $2x + y + z = 11$. Also, identify basic and non-basic variables.	2	3	2
	PART-B			
01.	Given the values of $u(x,y)$ on the boundary of the square region. Evaluate the function $u(x,y)$ satisfying Laplace equation $\nabla^2 u = 0$ at the pivotal points of this figure. Perform 6 iterations by Liebman method. 	10	1	2

02.	Determine the values of $u(x,t)$ satisfying the parabolic equation $\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}$ and the boundary conditions $u(0,t) = u(8,t) = 0$ and $u(x,0) = 4x - \frac{x^2}{2}$ at the points $x = i : i = 0,1,2,\ldots,8$ and $t = \frac{j}{8} : \ j = 0,1,2,\ldots,5$.	10	3	1
03.	The transverse displacement u of a point at a distance x from one end and at any time t of a vibrating string satisfies the equation $16\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial t^2} = 0$, with boundary conditions $u(0,t) = u(5,t) = 0$ and the initial conditions $u(x,0) = \begin{cases} 3x, & 0 \le x \le \frac{5}{2} \\ 20 - 4x, & \frac{5}{2} \le x \le 5 \end{cases}$ and $\frac{\partial u}{\partial t} = 0$ at t = 0. Solve the equation numerically up to t = 1, taking h = 1 and k = 1/4.	10	2	2
04.	The temperature distribution $u(x,t)$ is modelled as the one dimensional heat equation $16\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $0 \le x \le 1, t > 0$ subject to the boundary conditions $u(0,t) = u(1,t) = 0$, and the initial condition $u(x,0) = 100(x - x^3)$. Determine the temperature distribution $u(x,t)$ by taking $\Delta x = h = \frac{1}{5}$ *and* $\Delta t = k = \frac{2}{25}$ using Schmidt relation for 5 levels.	10	2	3
05.	A factory uses three types of machines to produce two types of electronic gadgets. The first gadget requires 12, 4 and 2 hours and the second gadget require 6, 10 and 3 hours respectively on the three types of machines. The total available time in hours on the three machines are 6000, 4000 and 1800. If the two types of gadgets fetch a profit of ₹400 and ₹1000, formulate the data as a linear programming problem and find the number of gadgets of each type to be produced for getting maximum profit using graphical method.	10	3	4

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	10	20	20	-	10	20	10	10	-	-
	Quiz	1	7	2	-	3	5	2	-	-	-

Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi

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Academic year 2022-2023 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**21st February 2023**	**Maximum Marks**	**10 + 50**
Course Code	**21IM32**	**Duration**	**20 + 90 Min**
Semester	**III Semester**	**CIE-II**	

MECHANICS OF MATERIALS

Note:
1. Answer all the Questions.

Sl. No.	PART - A Questions	M	BT	CO
1.	The number of phases present in Equilibrium at Eutectic point is _____	1	3	1
2.	What reaction does this equation denote? Liquid + Solid1 →Solid 2	1	2	2
3.	Expand T T T Diagram?	1	2	3
4.	What is the triple point of water?	2	2	1
5.	Enumerate Gibb's Phase rule?	2	3	3
6.	α- ferrite has a _____ crystal structure	1	2	2
7.	What is the composition of Carbon at cementite?	1	2	3
8.	For a circular shaft of diameter subjected to torque the maximum value of the shear stress is _____	1	2	2

Sl. No.	PART - B Questions	M	BT	CO
1.	Derive Torsion Equation with a neat diagram	10	3	2
2.	Draw Fe-Fe3C diagram and differentiate between cementite, austrinite and ferrite.	10	2	3
3.	A solid shaft rotating at 500 rpm transmits 30 KW. Max torque is 20% more than the mean torque. Material of the shaft has an allowable shear stress 65 Mpa and modulus of rigidity 81 GPa. Angle of twist in the shaft should not exceed 10 per meter length of shaft. Determine the safe diameter of shaft.	10	2	3
4.	Two metals Al and Cu have melting point at 800 °C and 500°C. The alloy pair forms eutectic at 400 °C of compositions 60%AL and 40% Cu. , the solid solubility are as follows: 20% Cu in Al at 400 °C and 10% Cu in Al at 0 °C 10% Al in Cu at 400 °C and 5% Al in Cu at 0 °C Assume the liquisus, solidus and solvent lines to be straight, no sold state reaction in the series i. Draw the phase diagram for the series and label all the salient temperature, composition and regions. ii. Find the amount of solid and liquid present at 450 °C	10	2	1
5.	Explain using cooling curves, how the Binary phase diagram are constructed?	10	3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	10	20	10	--	--	30	20	--	--	--
	Quiz		03	03	04	--	--	--	07	03	--	--	--

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**22nd February 2023**	**Maximum Marks**	**10 + 50**
Course Code	**21IM33**	**Duration**	**20 + 90 Min**
Sem	**III Semester**	**CIE – II**	

WORK SYSTEMS DESIGN

Note:
1. Answer all the Questions.

	PART - A			
Sl. No.	**Questions**	**M**	**BT**	**CO**
1	Distinguish between snap back and cumulative timing methods.	02	2	3
2	According to Taiichi Ohno, the number of waste forms in a manufacturing environment is _____.	01	1	2
3	The meaning of the word Muda in Japanese is _____.	01	1	2
4	Distinguish between accuracy and precision.	02	2	3
5	Distinguish between First generation and Higher generation levels of PMTS.	02	2	3
6	List the five steps in the direct time study procedure.	02	1	3

	PART - B			
Sl. No.	**Questions**	**M**	**BT**	**CO**
1.	Classify and explain briefly different methods used to determine the time standards.	10	3	3
2.	What is a Work Element? Mention at least six guidelines for defining work elements in direct time study. Supplement with appropriate explanation and examples.	10	3	3
3.	The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element *c* is a machine controlled and the time is constant. Elements *a,b,d,e* and *f* are operator controlled and external to machine cycle and they are performance rated at 80%.If the machine allowance is 25% and worker PFD allowance is 15%, determine the normal time and standard time for the work cycle.	10	3	3
4.	Enumerate the three activity sequence models in Basic MOST with suitable examples.	10	4	3
5.	Illustrate the structure of Lean Production System. Explain the features of different elements of the structure.	10	3	2

Element	a	b	c	d	e	f
Observed Time (min)	0.18	0.30	0.88	1.12	1.55	1.80

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	**Particulars**		**CO1**	**CO2**	**CO3**	**CO4**	**CO5**	**L1**	**L2**	**L3**	**L4**	**L5**	**L6**
	Test	**Max Marks**	--	10	40	--	--	--	--	40	10	--	--
	Quiz		--	06	04	--	--	04	06	--	--	--	--